

ECE 447

Lesson 20

PCM (cont'd) and Pulse Modulations



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **HW4.** Due Lesson 21 (Tue, 6 Oct) to Gradescope.
 - **HW5.** Assigned Lesson 21 (Tue, 6 Oct); due Lesson 23 (Tue, 13 Oct).
 - **Lab 5.** Begins Lesson 21 (Tue, 6 Oct); due Lesson 24 (Thu, 15 Oct).

PCM COMPUTER EXAMPLE

- What has a larger impact on quantization error: sampling frequency or number of bits?

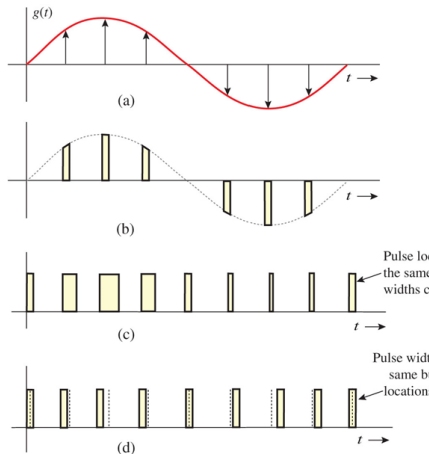
BINARY PCM

- SNR
 - Rule of thumb: each additional bit adds 6dB of SNR (how many times greater is that?)
- Max information rate given finite bandwidth
 - Assuming a noiseless channel with bandwidth B , a max of $2B$ pieces of info (e.g., bits) per second can be transmitted error-free
 - Shannon-Hartley theorem defines limit in presence of noise
- Transmission Bandwidth
 - For $m(t)$ band-limited to B Hz, min sample rate is $2B$ samples/s
 - Encoder output is then $2nB$ bit/s
 - Minimum required (theoretical) channel bandwidth,

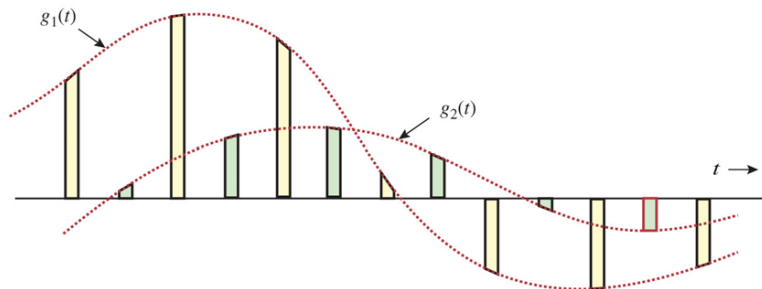
$$B_T =$$

PULSE MODULATIONS

- **PAM:** primarily now used for baseband digital communications (e.g., USB4 Ver 2.0 uses PAM-3, or 3-level PAM)
- **PWM:** power or motor control (e.g., robotics, power supply regulation); minimal telecommunication applications
- **PPM:** fiber optical communications, deep-space communications, R/C applications



TIME DIVISION MULTIPLEXING



BINARY PCM - EXAMPLE PROBLEM 5.2

- $m(t)$ band-limited to 3kHz is sampled at 1/3 higher than the Nyquist rate
- Max acceptable quantization error is 0.5% m_p
- Find min channel bandwidth for transmission.
- If 24 signals combined via TDM, what is min required transmission bandwidth of multiplexed signal?